INFOMSCIP 2026-2027
Assignment 4: Statistical Analysis of User Data

General comments: In this study you will work on a real open access data set published in Frontiers in Virtual Reality. In the publication, the authors already perform an analysis on the data. In this assignment we will think about the choices made with respect to data collection and analysis, and discuss how other choices may have affected the analysis and conclusions. You are expected to hand in a report, but you do not have to hand in your entire R scripts (although some questions ask for specific in and output).

General rules: Send your final group report to me by email (a.elali@uu.nl) as a PDF.

The following formalities should be considered:

- Download the zip package 'acmart-primary.zip' from the Assignments page on Brightspace, and use sample-sigconf.tex. Check 'latex_template_readme.pdf' for instructions.
- For the PDF file, use the name INFOMSCIP-X-Y.pdf where X is the number of the assignment and Y is your group number. Put all group members' names and student IDs at the beginning of the PDF and report here the time spent by each group member separately. You can do this in the author fields by presenting your name as e.g., *Abdo El Ali - 7###### - 10 hours*.

Use of Generative AI / LLMs for writing is not allowed: Please avoid using LLMs for writing in this (or future) assignments. Three reasons: (1) while LLM-generated language may look readable and plausible, the resulting work is often shallow, generic, or insufficiently grounded, which may be apparent during assessment; (2) while such tools can be useful for many tasks, relying on them for assignments like this can reduce your own learning and development. This applies not only to this course, but across your Master's program, and ultimately defeats the purpose of completing Master's-level work; and (3) submissions may be screened using AI-text detection tools, including Pangram, GPTZero, etc. Detector results are indicators and may lead to further review by the examination committee. Please do not use AI tools for data analysis, this defeats the purpose of your learning for this assignment.

Deadline for this assignment: Submit the final version of the PDF before **Thursday, October 15, 2026, 13:00**. Deadlines are strict. Each additional day after the deadline will result in a grade deduction of 1.0.

PART I: *Preparation.*

For this assignment we will be using R as our statistical computing language. As students of the UU, you can install R for Windows version 4.1.2 for free through SolisWorkplace. If you are using a different operating system than Windows, work together with a group member who does. For those of you who are unfamiliar with R, it is a relatively straightforward programming language, and there are many tutorials out there to learn the basics. I recommend using RStudio as development environment, especially if you are new to R and data analysis, but it is not required. RStudio is also available to you for free through SolisWorkplace. We will be using the default packages like `base` and `stats`, and for one question you are allowed to use other packages if desired. RStudio has a panel for installing and loading packages, usually on the bottom right, tab Packages.

The data we will be using for this assignment is from the article:

Pohl H and Mottelson A (2022) Hafnia Hands: A Multi-Skin Hand Texture Resource for Virtual Reality Research. Front. Virtual Real. 3:719506. doi: 10.3389/frvir.2022.719506

https://www.frontiersin.org/articles/10.3389/frvir.2022.719506/full

It is an open access article in the Frontiers in Virtual Reality journal. Read the article. For this assignment, we shall use the authors' data set that is linked under Section Data Availability Statement of the article. We will only use the csv's Demographics and Responses. Load the data in R.

Before we do any testing on the data, answer the following questions from the paper:

1. In your *own* words (please no LLM!), what is the motivation of the paper?

2. What are the contributions (to the scientific community) of the paper?

3. Which research contribution types introduced by Wobbrock and Kientz (2016) for HCI apply to this paper? (Max 2)

> **To deliver:** Answers to above questions, each in a few sentences (no more than 5!). You do not need to restate each question, just use sections for each Part, and subsections for each Question. The answers should be written full-out (so no bullet points) as you would expect in a paper, unless stated otherwise.

PART II: *Dealing with repetitions*

Let us focus on only Body Ownership for now. The authors state to have used a Friedman test to examine the Body Ownership responses. The Friedman test is similar to the one-way ANOVA we discussed in the lecture. It is a non-parametric test for a repeated measures design with a factor that has three or more levels, or in other words, it is used when we have data of unknown (or a non-parametric) distribution in a within-subjects design with at least three conditions for only one independent variable. The authors' response data is ordinal of an unknown distribution, and the design was indeed within-subjects, since each subject used three different hand representations. However, the authors also state that each participant used each hand three times. These are called repetitions (not related to the 'repeated measures' design mentioned earlier), and these repetitions can have different response values. This is true for our data set: for example participant 9 responded to Body Ownership question with a 2 – 0 – 2 in the repetitions of the Matched Hand condition. Since a Friedman test does not allow for repetitions, we must assume the authors first transformed the data by combining the three values into a single value before analysis. To be clear, the inclusion of repetitions technically requires a completely different test than a 'plain' Friedman test. If the authors have reason to use a different approach, they should have reported this. Today, we shall cut them some slack and try to figure out what they did.

1. What is a possible drawback to using a single combined value of the repetition values as opposed to performing a different test that retains the repetition values, and how could this affect the conclusions of the experiment? You can use citations to back up your response.

2. Looking at the different measures of central tendency discussed in the slides, name one measure that could be used for this data set to compute a combined response value, and explain why it could be a valid measure to use? There are multiple valid answers here; pick the one you think the reviewers of the article would complain the least about. Also, explain which measure would be invalid and explain why it would be invalid.

3. Use the `friedman.test` from the `stats` package to perform a Friedman test on this Body Ownership data, to see if it matches the authors' results. First, create a new smaller dataset that contains values for 'PID', 'Condition' and 'Response'. The latter column should contain the newly computed combined body ownership values according to the measure of central tendency you chose in the previous question. If the effect is significant, then also

perform pairwise comparisons to see which conditions actually differ significantly from each other, using the `wilcox.test` for Wilcoxon signed rank tests from the `stats` package. Remember that since you are performing multiple comparisons, you must perform some type of correction to account for the increased likelihood of a Type I error. Do this by hand after computing the results of the Wilcoxon signed rank test using a Bonferroni correction. You do not need to calculate power.

To deliver: Answers to the first two questions above, each in a few sentences (not more than 5). For the third question, list the exact R input (i.e., code) and output of the `friedman.test` and, if applicable, the exact R input and output of the pairwise `wilcox.test`. State whether they match the results in the article, and if not, give a possible explanation. You do not need any packages besides those loaded by default, but if you do choose to use some, then list them. Make sure to correctly and clearly report your results and conclusions as you would in a paper. This includes clearly and correctly reporting your statistical results; if you are uncertain how to report this, then google "how to report Friedman test results" (this is how you will approach this problem in your future studies as well). For the R input and output, copy paste or add a *legible* screenshot of the text in R. Make sure to give your datasets and dataframes appropriate names. You do not need to include the dataframes themselves in the PDF.

PART III: *Differences between participants*

The demographics data reveal many ways in which the participants differed from each other. In the Validation section and Limitations section, the authors discuss how sex and skin tone distributions could have impacted their results.

1. The authors used multiple Fisher's exact tests to examine whether there was an effect of sex on any of the dependent variables (independently from each other). A Fisher's exact test is typically used to examine association between two categorical variables, when both variables only have two options. This is not the case, since the responses are on 7-point scales, and have natural order. Give an example of what way the authors could have created categorical data from the response data. There are multiple valid answers here. Pick the one you think the reviewers of the article would complain the least about, and explain why it is valid for the particular Body Ownership question.

2. Again taking the body ownership data as an example, describe a statistical test that you think is better suited to examine the effect

of sex (just Male & Female), regardless of hand representation. Use the links provided in the lecture slides to choose a test, and if you are not sure, google extra information about possible tests (as this is probably how you will approach this problem in the future). State the null and alternative hypotheses, and for the latter, explicitly state whether it is one-sided or two-sided. List the variables and corresponding levels, and all parameters you need in your test. Be as specific as possible. You do not need to actually perform this test in R.

Compared to previous virtual hand illusion experiments, for example around 2000-2015, this experiment had an unusually large portion of the participants with VR experience; almost half had more than 100 hours of experience! We informally argue that participants who have used VR for longer are more likely to be acquainted with the concept of body ownership, and are therefore more likely to be able to judge what is being tested and know the "correct" answer. They then (subconsciously) answer correspondingly in a favorable way, even though it may not reflect their current experience. This is a form of demand bias. We posit a positive correlation between the amount of VR experience and body ownership responses.

3. Using the data set you created in Part II as a starting point (that is, where each condition only has a single response value rather than three repetitions), visualize the *posited* correlation between amount of VR experience and body ownership responses for each hand representation, i.e. choose a visualization, and create one plot for each condition. By *posited* I mean that you do not actually have to calculate the correlation or draw a correlation function (see next question), just find a way to visualize the data in such way that would indicate to readers they should admit or rule out a possible correlation. Choose what you think is the best visualization based on the type of data we are using, the variables we are investigating, and the problem we are trying to investigate. You will have to make decisions here, since there is no perfect answer. Make sure your figures contain all the information a reader of a paper would need to understand it. Feel free to experiment with plotting functions from other R packages than the default packages.

4. Test post-hoc whether there is a positive correlation between VR experience and body ownership responses for each hand representation separately, meaning you will have to do multiple comparisons. Choose the correct correlation test using the links in the slides and by googling if necessary. Report all relevant results and draw a conclusion from your results. If you do not understand

how to interpret the results in order to give a conclusion, google how to interpret the values of your chosen correlation test. You do not need to calculate power.

To deliver: Answer to the first question in a few sentences (1-3 max). Answer to the second question in 1-2 paragraphs. For the third question, provide the visualizations with correct formatting and descriptive captioning. Think about figure presentation, as we covered in one of the earlier lectures. If you used packages other than the default packages (e.g., ggplot2, seaborn), list these other packages. For the fourth question, explain your choice of correlation test, list your exact R input (i.e., code) and output. You do not need any packages besides those loaded by default, but if you do choose to use some, then list them. Make sure to correctly and clearly report your results and conclusions as you would in a paper. This includes clearly and correctly reporting your statistical results.